```
TERMINAL
➜ 26 sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 16
Server version: 8.0.44-0ubuntu0.22.04.2 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input stat

mysql> SOURCE sql_databasesql-d5pqoasrtl0hpva4650g.sql;
```

```
mysql> SOURCE sql_databasesql-d5pqoasrtl0hpva4650g.sql;
Database changed
Query OK, 0 rows affected (0.00 sec)

Query OK, 0 rows affected (0.40 sec)

Query OK, 0 rows affected (0.16 sec)

Query OK, 0 rows affected (0.18 sec)
```

```
Query OK, 24360 rows affected (0.42 sec)
Records: 24360  Duplicates: 0  Warnings: 0

Query OK, 24360 rows affected (0.31 sec)
Records: 24360  Duplicates: 0  Warnings: 0

Query OK, 12609 rows affected (0.16 sec)
Records: 12609  Duplicates: 0  Warnings: 0

Query OK, 21707 rows affected (0.38 sec)
Records: 21707  Duplicates: 0  Warnings: 0

Query OK, 21700 rows affected (0.20 sec)
Records: 21700  Duplicates: 0  Warnings: 0

Query OK, 21687 rows affected (0.21 sec)
Records: 21687  Duplicates: 0  Warnings: 0

Query OK, 21694 rows affected (0.25 sec)
Records: 21694  Duplicates: 0  Warnings: 0

Query OK, 21695 rows affected (0.28 sec)
Records: 21695  Duplicates: 0  Warnings: 0

Query OK, 21700 rows affected (0.20 sec)
Records: 21700  Duplicates: 0  Warnings: 0

Query OK, 21326 rows affected (0.23 sec)
```

```
Records: 24921  Duplicates: 0  Warnings: 0

Query OK, 24922 rows affected (0.17 sec)
Records: 24922  Duplicates: 0  Warnings: 0

mysql>
```

SQL Questions for assignment:

1.Find the 2nd highest salaried person in the finance department and also find his name?

```
SELECT e.emp_no,e.first_name,e.last_name
FROM employees e
WHERE e.emp_no=(
SELECT emp_no FROM salaries
WHERE emp_no IN(
SELECT emp_no FROM dept_emp
```

```
WHERE dept_no=(SELECT dept_no FROM departments WHERE dept_name='Finance')
)
AND salary<(
SELECT MAX(salary) FROM salaries
WHERE emp_no IN(
SELECT emp_no FROM dept_emp
WHERE dept_no=(SELECT dept_no FROM departments WHERE dept_name='Finance')
)
)
ORDER BY salary DESC LIMIT 1
);
```

```
mysql> SOURCE Gurivi_reddy_yasaswini_Subquery.sql;
+------------+
|            |
+------------+
| exercise 1 |
+------------+
1 row in set (0.00 sec)

+--------+------------+-----------+
| emp_no | first_name | last_name |
+--------+------------+-----------+
| 109084 | Jiang      | Vesel     |
+--------+------------+-----------+
1 row in set (0.10 sec)
```

2.Fetch the list of of employees who worked in Production department between 1998 and 2000?

```
select "exercise 2"  as '';
SELECT DISTINCT e.emp_no,e.first_name,e.last_name
FROM employees e
WHERE e.emp_no IN(
  SELECT emp_no
  FROM dept_emp
  WHERE dept_no=(SELECT dept_no FROM departments WHERE dept_name='Production')
  AND from_date<='2000-12-31'
  AND to_date>='1998-01-01'
)
```

```
|            |
+------------+
| exercise 2 |
+------------+
1 row in set (0.00 sec)

+--------+------------+-----------+
| emp_no | first_name | last_name |
+--------+------------+-----------+
|  10003 | Parto      | Bamford   |
|  10004 | Chirstian  | Koblick   |
|  10010 | Duangkaew  | Piveteau  |
|  10018 | Kazuhide   | Peha      |
|  10020 | Mayuko     | Warwick   |
|  10024 | Suzette    | Pettey    |
|  10026 | Yongqiao   | Berztiss  |
|  10029 | Otmar      | Herbst    |
```

```
TERMINAL
| 495331 | Haldon        | Taubenfeld     |
| 495334 | Shuky         | Gajiwala       |
| 495335 | Persi         | Rubsam         |
| 495338 | Rildo         | Brodal         |
| 495342 | Shigenori     | Cooke          |
| 495353 | Odysseas      | Tischendorf    |
| 495355 | Georgi        | Panangaden     |
| 495357 | Izaskun       | Chaudhuri      |
| 495358 | Giordano      | Schnabel       |
| 495360 | Djenana       | Christ         |
| 495362 | Sushant       | Raoux          |
| 495363 | Hatsukazu     | Narahari       |
| 495364 | Sasan         | Jeansoulin     |
| 495369 | Kersti        | Bressoud       |
| 495371 | Koldo         | Cherinka       |
| 495376 | Charlene      | Collette       |
| 495381 | Munir         | Cooley         |
| 495385 | Ramachenga    | Benzaken       |
| 495393 | Vugranam      | Perko          |
| 495395 | Nectarios     | Emiris         |
| 495396 | Jinya         | Krupka         |
| 495398 | Bingning      | Yetto          |
| 495401 | Eishiro       | Bernick        |
| 495404 | Abdulah       | Merli          |
```

3.Who is the 3rd highest salaried person in Production department without using orderby.

```
SELECT 'exercise 3' AS '';
SELECT e.emp_no,e.first_name,e.last_name
FROM employees e
WHERE e.emp_no IN(
SELECT s.emp_no
FROM salaries s
WHERE s.to_date='9999-01-01'
AND s.salary=(
SELECT MAX(salary)
FROM salaries
WHERE to_date='9999-01-01'
AND emp_no IN(
SELECT emp_no FROM dept_emp
WHERE dept_no=(SELECT dept_no FROM departments WHERE dept_name='Production')
)
AND salary<(
SELECT MAX(salary)
```

```
FROM salaries
WHERE to_date='9999-01-01'
AND emp_no IN(
SELECT emp_no FROM dept_emp
WHERE dept_no=(SELECT dept_no FROM departments WHERE dept_name='Production')
)
AND salary<(
SELECT MAX(salary)
FROM salaries
WHERE to_date='9999-01-01'
AND emp_no IN(
SELECT emp_no FROM dept_emp
WHERE dept_no=(SELECT dept_no FROM departments WHERE dept_name='Production')
)
)
)
)
);
```

```
+------------+
|            |
+------------+
| exercise 3 |
+------------+
1 row in set (0.00 sec)

+--------+------------+-----------+
| emp_no | first_name | last_name |
+--------+------------+-----------+
|  61036 | Shimshon   | Nergos    |
+--------+------------+-----------+
1 row in set (0.90 sec)

mysql>
```

4.Find how many employees joined in which year in a complete single report.

```
select "exercise 4"  as '';

SELECT YEAR(hire_date),COUNT(*)
FROM employees
GROUP BY YEAR(hire_date);
```

```
TERMINAL
+------------+
| exercise 4 |
+------------+
1 row in set (0.00 sec)

+-----------------+----------+
| YEAR(hire_date) | COUNT(*) |
+-----------------+----------+
|            1986 |    36150 |
|            1985 |    35316 |
|            1989 |    28394 |
|            1994 |    14835 |
|            1990 |    25610 |
|            1992 |    20402 |
|            1987 |    33501 |
|            1995 |    12115 |
|            1993 |    17772 |
|            1999 |     1514 |
|            1991 |    22568 |
|            1988 |    31436 |
|            1997 |     6669 |
|            1996 |     9574 |
|            1998 |     4155 |
|            2000 |       13 |
+-----------------+----------+
16 rows in set (0.07 sec)
```

5.Fetch the maximum employees joined year from the above report without using having.

```
select "exercise 5"  as '';

SELECT YEAR(hire_date)
FROM employees
GROUP BY YEAR(hire_date)
ORDER BY COUNT(*) DESC LIMIT 1;
```

```
+------------+
|            |
+------------+
| exercise 5 |
+------------+
1 row in set (0.00 sec)

+-----------------+
| YEAR(hire_date) |
+-----------------+
|            1986 |
+-----------------+
1 row in set (0.07 sec)
```

6.Need the reports total cost divided yearly in a table of all departments.

```
select "exercise 6"  as '';
```

```
SELECT d.dept_name,YEAR(s.from_date),SUM(s.salary)
FROM salaries s
JOIN dept_emp de ON s.emp_no=de.emp_no
JOIN departments d ON de.dept_no=d.dept_no
WHERE s.from_date BETWEEN de.from_date AND de.to_date
GROUP BY d.dept_name,YEAR(s.from_date);
```

```
|            |
+------------+
| exercise 6 |
+------------+
1 row in set (0.00 sec)

+--------------------+-------------------+---------------+
| dept_name          | YEAR(s.from_date) | SUM(s.salary) |
+--------------------+-------------------+---------------+
| Customer Service   |              1985 |      21622993 |
| Customer Service   |              1986 |      47369131 |
| Customer Service   |              1987 |      73010083 |
| Customer Service   |              1988 |     101352441 |
| Customer Service   |              1989 |     130030259 |
| Customer Service   |              1990 |     159320515 |
| Customer Service   |              1991 |     189543590 |
| Customer Service   |              1992 |     221596131 |
| Customer Service   |              1993 |     251110669 |
| Customer Service   |              1994 |     282421834 |
| Customer Service   |              1995 |     312254799 |
| Customer Service   |              1996 |     344411025 |
| Customer Service   |              1997 |     374708287 |
| Customer Service   |              1998 |     404098974 |
| Customer Service   |              1999 |     431073691 |
| Customer Service   |              2000 |     437843946 |
| Customer Service   |              2001 |     437424703 |
| Customer Service   |              2002 |     254269912 |
| Development        |              1985 |      86021286 |
| Development        |              1986 |     181040513 |
| Development        |              1987 |     281840161 |
| Development        |              1988 |     385090525 |
| Development        |              1989 |     490720968 |
| Development        |              1990 |     598966251 |
| Development        |              1991 |     704631037 |
| Development        |              1992 |     819886874 |
| Development        |              1993 |     932397380 |
| Development        |              1994 |    1037296241 |
| Development        |              1995 |    1147213106 |
| Development        |              1996 |    1261547637 |
| Development        |              1997 |    1368724600 |
| Development        |              1998 |    1471786689 |
| Development        |              1999 |    1573845119 |
```

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Development	1999	1573845119
Development	2000	1594365577
Development	2001	1594721310
Development	2002	936507545
Finance	1985	22381837
Finance	1986	44596157
Finance	1987	68773383
Finance	1988	93389938
Finance	1989	120035809
Finance	1990	146599434
Finance	1991	172639263
Finance	1992	199026100
Finance	1993	224286937
Finance	1994	249541547
Finance	1995	274617880
Finance	1996	300201712
Finance	1997	324780274
Finance	1998	348104580
Finance	1999	371274971
Finance	2000	373038800
Finance	2001	371457188
Finance	2002	219623933
Human Resources	1985	16204774
Human Resources	1986	34859232
Human Resources	1987	54474010
Human Resources	1988	73461301
Human Resources	1989	93388607
Human Resources	1990	112424713
Human Resources	1991	133362846
Human Resources	1992	154794903
Human Resources	1993	175705490
Human Resources	1994	199202321
Human Resources	1995	222697275
Human Resources	1996	245400027
Human Resources	1997	268950695
Human Resources	1998	289112190
Human Resources	1999	308664752
Human Resources	2000	312746359
Human Resources	2001	314454603
Human Resources	2002	184436853
Marketing	1985	27880162
Marketing	1986	55994427

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DEBUG CONSOLE PROBLEMS TERMINAL PORTS

TERMINAL

Production	1992	700662984
Production	1993	795709513
Production	1994	892745854
Production	1995	987160731
Production	1996	1084422492
Production	1997	1178043537
Production	1998	1270944500
Production	1999	1358022181
Production	2000	1378960945
Production	2001	1378373416
Production	2002	806331782
Quality Management	1985	19093581
Quality Management	1986	39096388
Quality Management	1987	62189241
Quality Management	1988	86078573
Quality Management	1989	107718626
Quality Management	1990	130932039
Quality Management	1991	156346447
Quality Management	1992	180268084
Quality Management	1993	205987596
Quality Management	1994	230608924
Quality Management	1995	255072460
Quality Management	1996	280844755
Quality Management	1997	305533180
Quality Management	1998	330559612
Quality Management	1999	352764608
Quality Management	2000	357655668
Quality Management	2001	358174605
Quality Management	2002	207016949
Research	1985	21779176
Research	1986	46814295
Research	1987	70171307
Research	1988	96009732
Research	1989	121535157
Research	1990	146928043
Research	1991	173330353
Research	1992	202865466
Research	1993	230181914
Research	1994	256460478
Research	1995	284475752
Research	1996	312737712

```
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| Quality Management |              2000 |      357655668 |
| Quality Management |              2001 |      358174605 |
| Quality Management |              2002 |      207016949 |
| Research           |              1985 |       21779176 |
| Research           |              1986 |       46814295 |
| Research           |              1987 |       70171307 |
| Research           |              1988 |       96009732 |
| Research           |              1989 |      121535157 |
| Research           |              1990 |      146928043 |
| Research           |              1991 |      173330353 |
| Research           |              1992 |      202865466 |
| Research           |              1993 |      230181914 |
| Research           |              1994 |      256460478 |
| Research           |              1995 |      284475752 |
| Research           |              1996 |      312737712 |
| Research           |              1997 |      339441781 |
| Research           |              1998 |      363994062 |
| Research           |              1999 |      389444419 |
| Research           |              2000 |      393890711 |
| Research           |              2001 |      395284078 |
| Research           |              2002 |      235438054 |
| Sales              |              1985 |       75491118 |
| Sales              |              1986 |      162331253 |
| Sales              |              1987 |      245201946 |
| Sales              |              1988 |      329999384 |
| Sales              |              1989 |      414262771 |
| Sales              |              1990 |      504243562 |
| Sales              |              1991 |      591473283 |
| Sales              |              1992 |      686334115 |
| Sales              |              1993 |      773988386 |
| Sales              |              1994 |      862705095 |
| Sales              |              1995 |      949735518 |
| Sales              |              1996 |     1038184028 |
| Sales              |              1997 |     1119794442 |
| Sales              |              1998 |     1206320322 |
| Sales              |              1999 |     1286002575 |
| Sales              |              2000 |     1292062186 |
| Sales              |              2001 |     1283939798 |
| Sales              |              2002 |      748321668 |
+--------------------+-------------------+----------------+
162 rows in set (3.02 sec)
```

7.Find the Year where highest female employees joined.

```
select "exercise 7"  as '';
SELECT YEAR(hire_date)
FROM employees
WHERE gender='F'
GROUP BY YEAR(hire_date)
ORDER BY COUNT(*) DESC LIMIT 1;
```

```
+------------+
|            |
+------------+
| exercise 7 |
+------------+
1 row in set (0.01 sec)

+-----------------+
| YEAR(hire_date) |
+-----------------+
|            1986 |
+-----------------+
1 row in set (0.08 sec)
```

8.Hire Date of the current manager of Human Resources department.(hint: to_Date year is 9999)

```
select "exercise 8"  as '';
SELECT hire_date
FROM employees
WHERE emp_no=(
SELECT emp_no FROM dept_manager
WHERE dept_no=(SELECT dept_no FROM departments WHERE dept_name='Human Resources')
AND to_date='9999-01-01'
);
```

```
+------------+
|            |
+------------+
| exercise 8 |
+------------+
1 row in set (0.00 sec)

+------------+
| hire_date  |
+------------+
| 1985-08-04 |
+------------+
```

9.Salary of the current manager of the Finance dept.

```
select "exercise 9"  as '';
SELECT salary
FROM salaries
WHERE emp_no=(
SELECT emp_no FROM dept_manager
WHERE dept_no=(SELECT dept_no FROM departments WHERE dept_name='Finance')
AND to_date='9999-01-01'
)
AND to_date='9999-01-01';
```

```
+------------+
|            |
+------------+
| exercise 9 |
+------------+
1 row in set (0.00 sec)

+--------+
| salary |
+--------+
|  83457 |
+--------+
```

10.Current dept managers first names along with emp_no.

```
-- -- Current department managers first name and emp_no
select "exercise 10"  as '';
SELECT e.first_name,e.emp_no
FROM employees e
WHERE e.emp_no IN(
SELECT emp_no FROM dept_manager
WHERE to_date='9999-01-01'
);
```

```
+-------------+
|             |
+-------------+
| exercise 10 |
+-------------+
1 row in set (0.00 sec)

+------------+--------+
| first_name | emp_no |
+------------+--------+
| Vishwani   | 110039 |
| Isamu      | 110114 |
| Karsten    | 110228 |
| Oscar      | 110420 |
| Leon       | 110567 |
| Dung       | 110854 |
| Hauke      | 111133 |
| Hilary     | 111534 |
| Yuchang    | 111939 |
+------------+--------+
9 rows in set (0.00 sec)
```